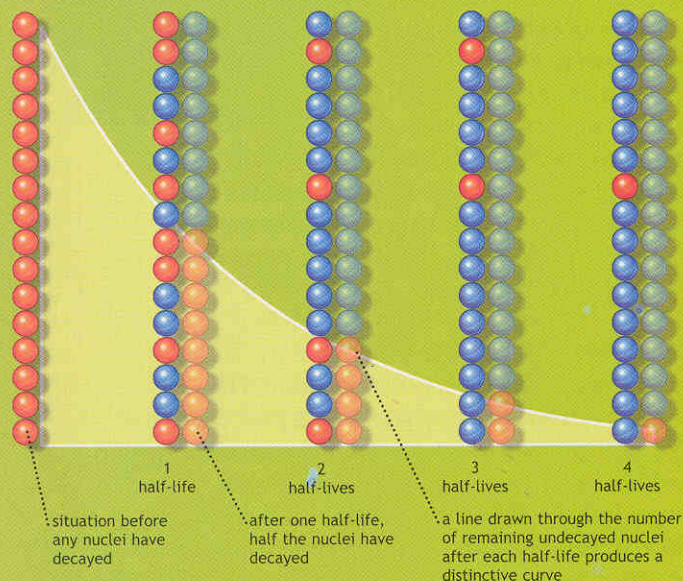


decay and half-life

When an individual atom (strictly speaking, a nucleus) undergoes radioactive decay, it does so in accordance with the rules of chance. It is as if the atom tossed a coin, and decayed if it came down heads, and didn't decay if it came down tails; or as if it rolled a die, and only decayed if an even number comes up. Then it makes

another toss, or roll. This means that in a large number of atoms, a certain proportion will always decay in a certain time. Physicists measure this in terms of the time it takes for half the atoms in a sample to decay, and they call this the half-life. Each kind of radioactive nucleus has its own distinctive half-life.



radioactive decay

The length of time covered by a half-life can range from less than a millionth of a second to millions of years depending on the element. The shorter the half-life, the more

radioactive the element is. For example, Nitrogen-16, which occurs in nuclear power stations has a half-life of seven seconds while Carbon-14, which is used in radio-carbon dating, has a half-life of 5730 years.

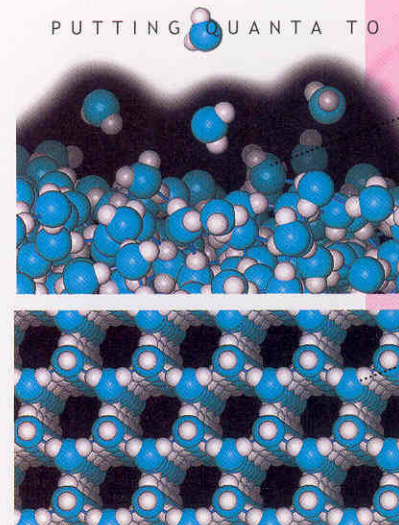
The result is that the positive charge on the proton is partly exposed to the outside world—but only partly exposed, because the wave function of the electron still extends around the proton, even though it is more strongly concentrated in the bond. In the classical view of electrons, this could not happen; the electron would have to

be on one side of the nucleus or the other. As a result of the partial shielding, hydrogen atoms that are already involved in chemical bonding on one side effectively show a fractional positive charge on the other side, and this can attract the electrons in a cloud around another

“When it comes to atoms, language can be used only as in poetry. The poet, too, is not nearly so concerned with describing facts as with creating images.”

Niels Bohr (1922)

can be calculated; it is about one-tenth as strong as a conventional bond, and this matches experimental measurements, again confirming that quantum physics works in the real world. But the key importance of the hydrogen bond is that it is what holds deoxyribonucleic acid (DNA) together. DNA, the molecule of life, is made of two intertwined strands (the famous double helix) held



in liquid water, molecules have looser bonds and evaporate from the surface

when frozen, water molecules crystallize into an open lattice

liquid and solid
In liquid water, permanent hydrogen bonds cannot form between the molecules because they possess too much energy and move too quickly. When water freezes, however, permanent hydrogen bonds lock into place, crystallizing the molecules in a very open lattice. This is why ice at 32°F is less dense than water, and why ice floats on water.